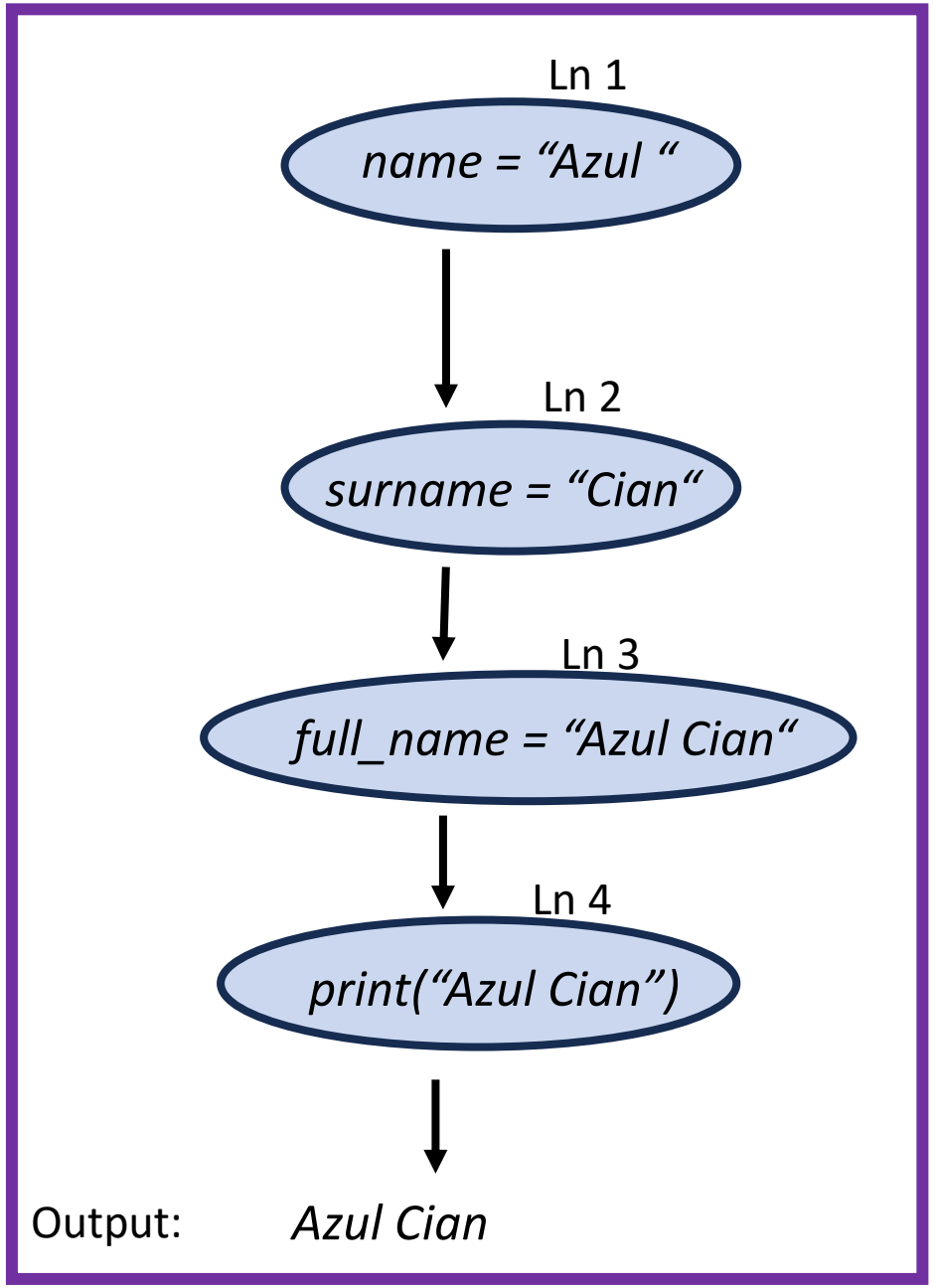
Lesson #7

Making Decisions!

(It just means "drawing where the code goes!")

This is a **CFG** (Control Flow Graph)



Up to now, our code has been "sequential"

```
1   name = "Azul "
2   surname = "Cian"
3   full_name = name + surname
4   print(full_name)
```

Do you remember the **Turing Machine**?

It has…

- 2 instructions
- 2 decisions

Making Decisions
Control Flow Structures / Branching

Turing Machine (1930)	"Binary" (ASM, 1947)	Present era (Python, 2026)
• **2** Decisions → *Left*, *Right*	• If (branch)	• if … else …
		• while …
• **2** Instructions → 0, 1	• Jump (go to)	• for … in …

Making Decisions
Control Flow Structures / Branching

Turing Machine (1930)	"Binary" (ASM, 1947)	Present era (Python, 2026)
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• 2 Instructions → 0, 1	• Jump (go to)	• for … in …

Present era

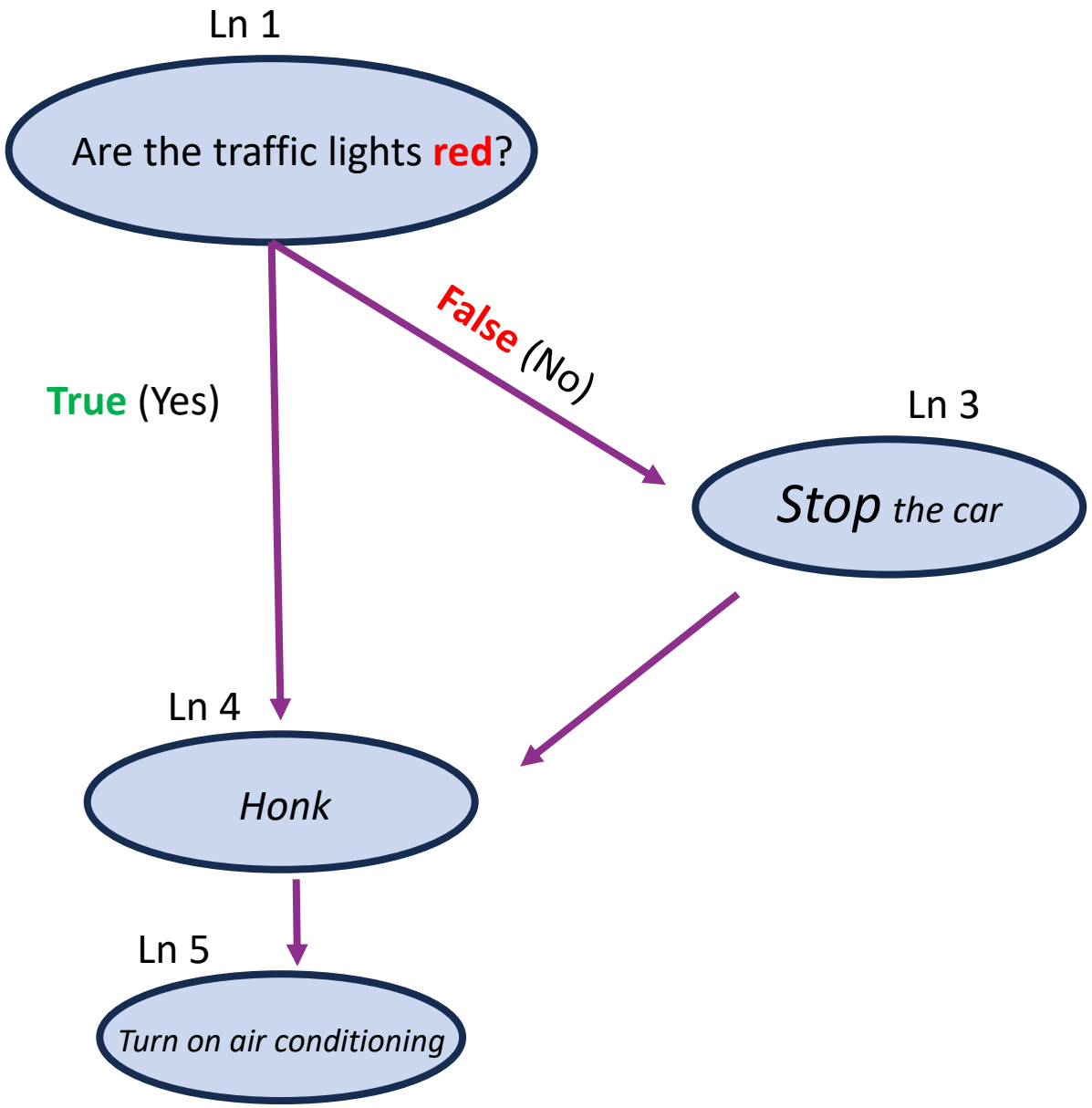
(Python, 2026)

- if BOOL else BOOL
- while BOOL
- for element in list

If

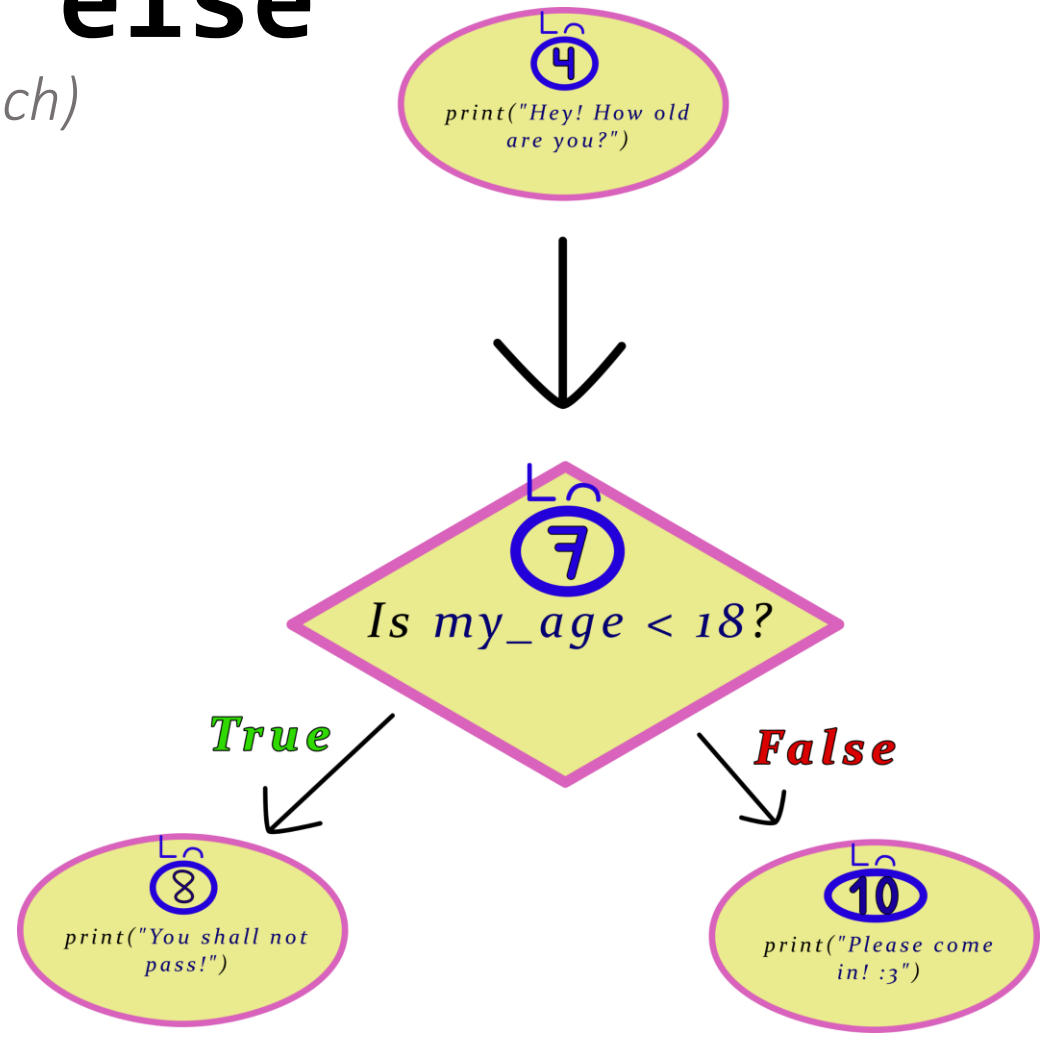
(branch)

```
10  if (traffic_lights == "red"):
11      stop_cars()
12
13  car_honk()
14  car_air_conditioner_on()
```



If else

(branch)

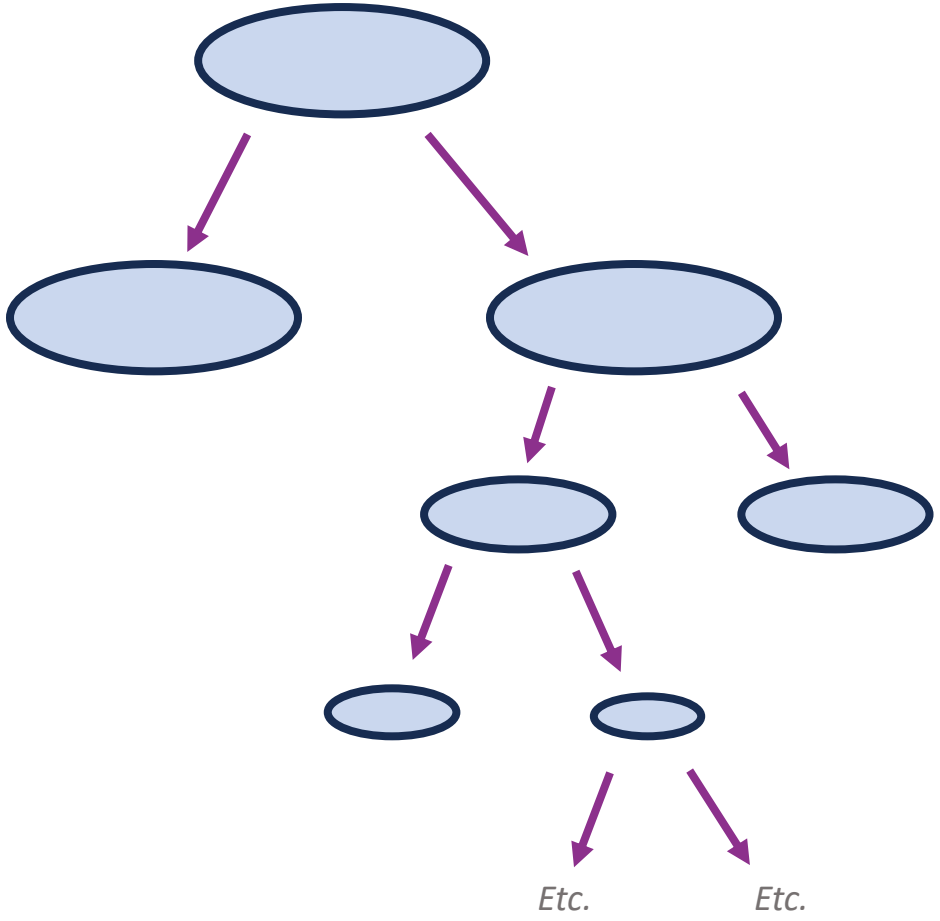



```python
# Disco example
# (Ejemplo boliche)

print("Hey! How old are you?\n")
my_age = input()

if (my_age < 18):
    print("You shall not pass!")
else:
    print("Please, come in! :3")
```

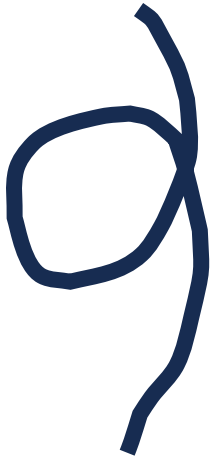
If elif elif elif elif… else

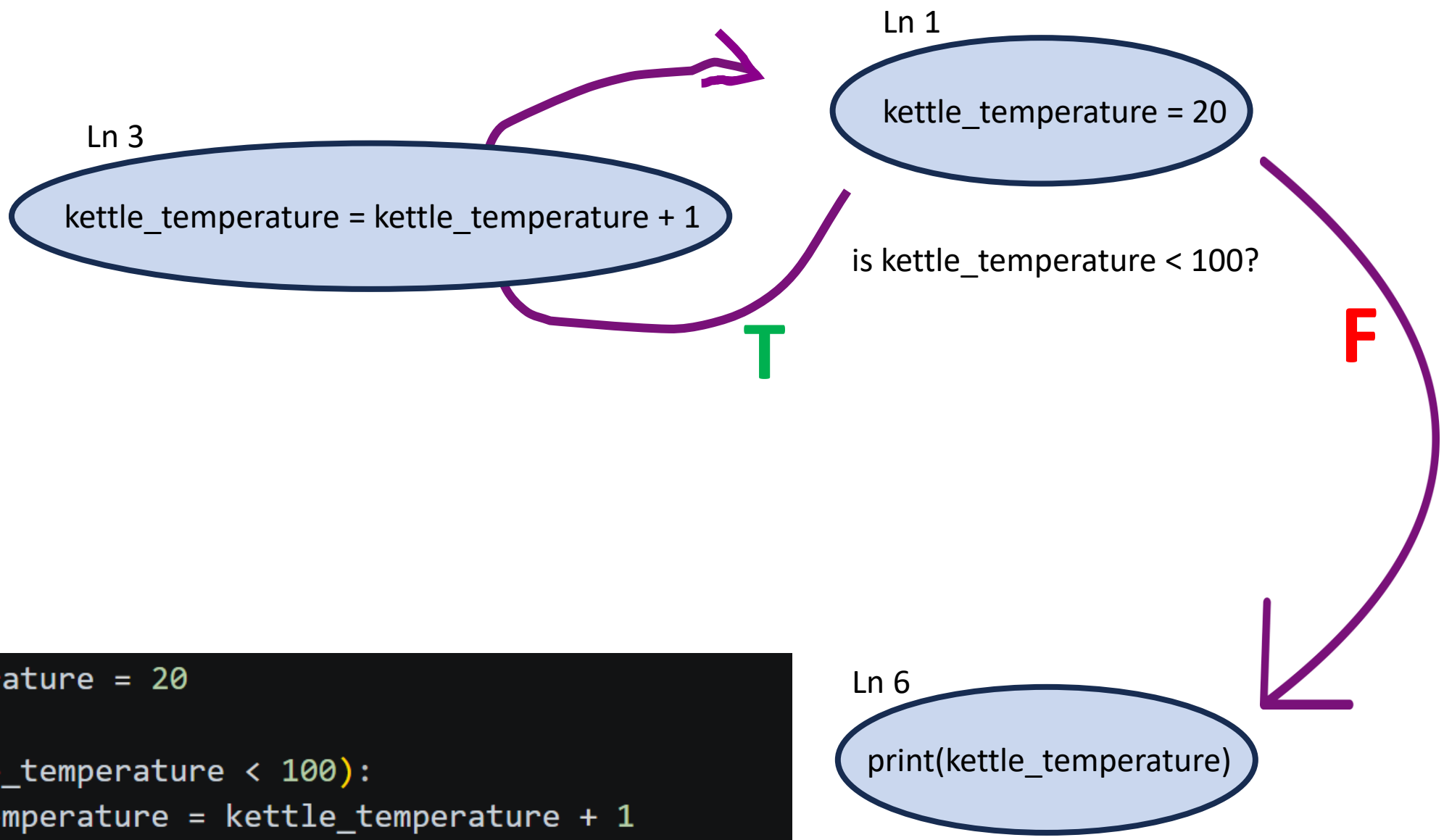


It’s very easy to *get lost*… right?

If you get lost when coding, do your own drawings on pen and paper!

While
(loop)

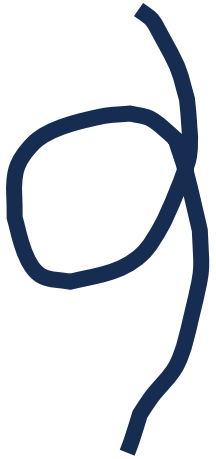


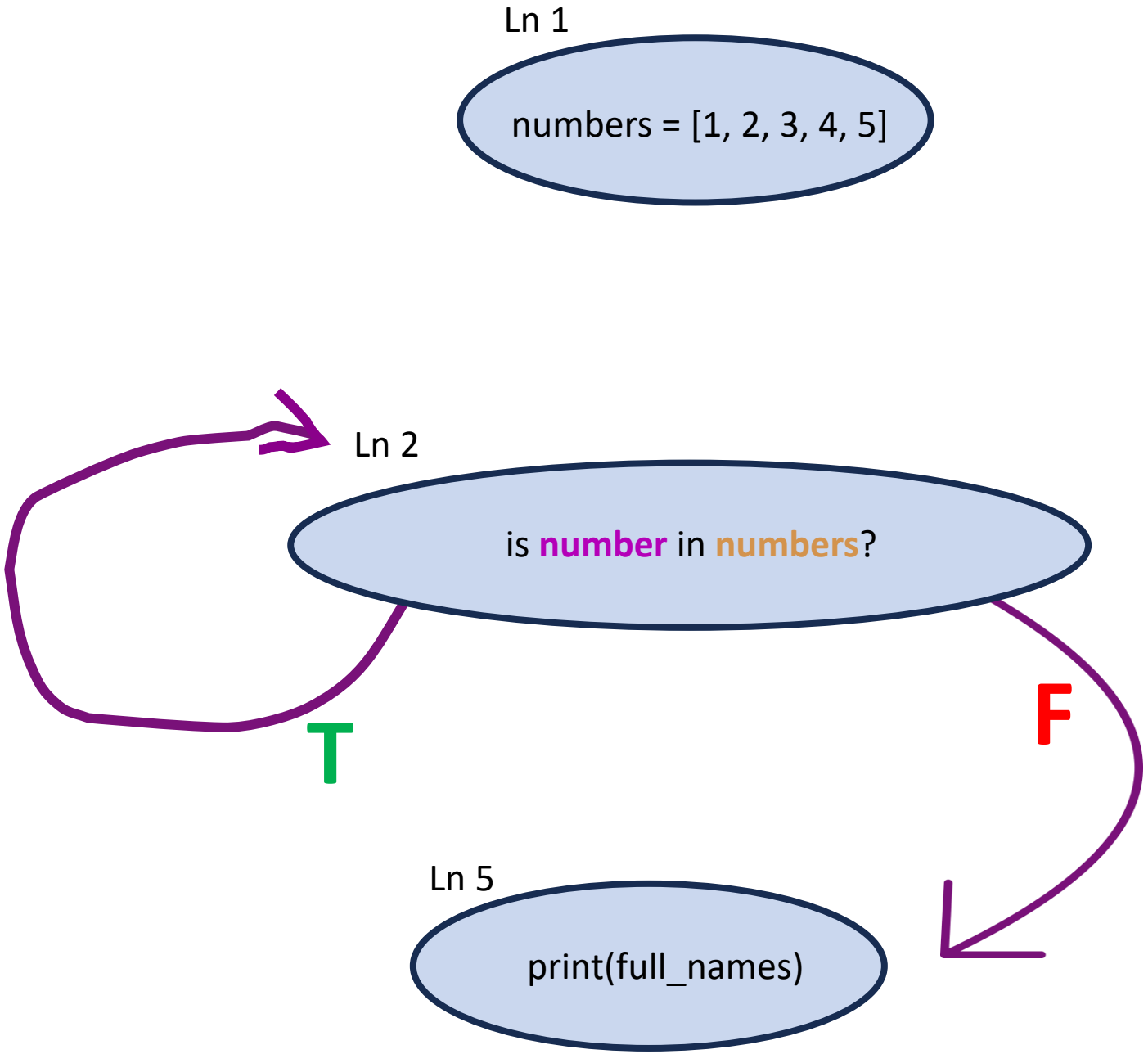


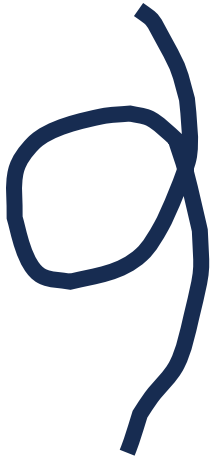
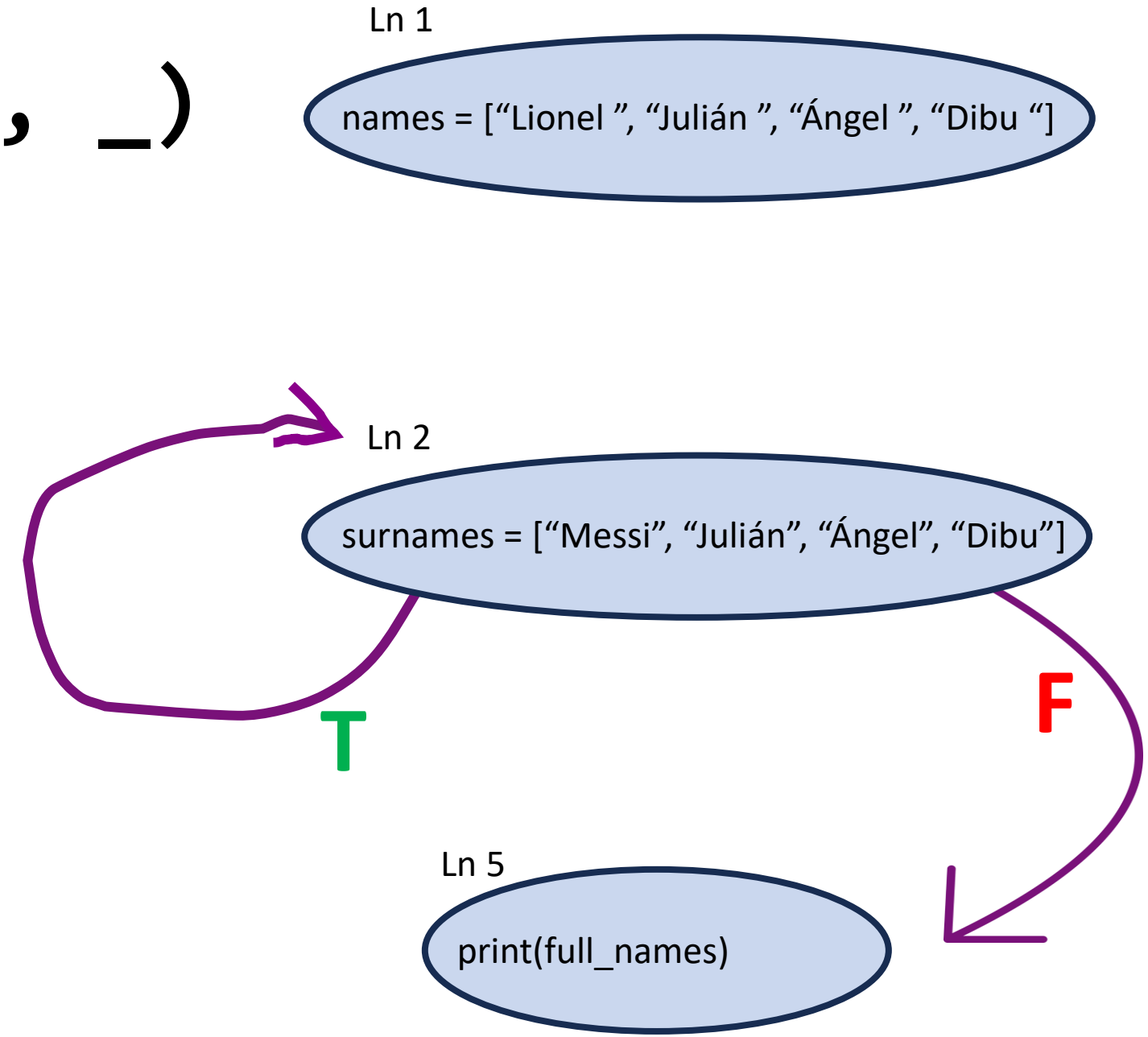

```
1   kettle_temperature = 20
2
3   while (kettle_temperature < 100):
4       kettle_temperature = kettle_temperature + 1
5
6   print(kettle_temperature)
```

For ____ in ____

(loop)




Ln 1
numbers = [1, 2, 3, 4, 5]
Ln 2
is number in numbers?
T
F
Ln 5
print(full_names)

For i in range(_, _)

(loop)